Class 18: Pertussis and the CMI-PB project

Nathalie Huang (PID: A19134713)

Table of contents

Background 1

CMI-PB project **8**

Focus on "PT" Pertusisis Toxin antigen 22

Obtaining CMI-PB RNASeq data 23

Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by the bacteria B. Pertussis.

This can infect all ages but is most severe for those under 1 year of age.

The CDC track the number of reported cases in US We can "scrape" this data with the **datapasta** package.

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
    1931L,
    1932L,
    1933L,
```

```
    1934L,
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
    1948L,
    1949L,
    1950L,
    1951L,
    1952L,
    1953L,
    1954L,
    1955L,
    1956L,
    1957L,
    1958L,
    1959L,
    1960L,
    1961L,
    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
    1973L,
    1974L,
    1975L,
```

```
    1976L,
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
    1992L,
    1993L,
    1994L,
    1995L,
    1996L,
    1997L,
    1998L,
    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
    2015L,
    2016L,
    2017L,
```

```r
    2018L,
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
    68687,
    45030,
    37129,
```

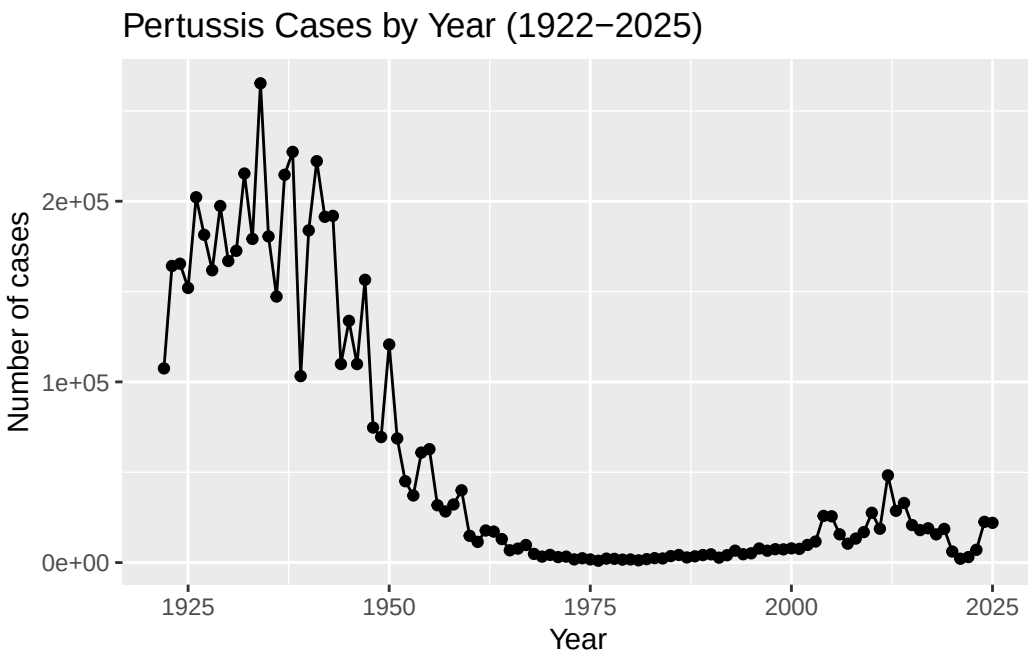
```
    60886,
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
    6586,
    4617,
    5137,
```

```
    7796,
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22538,
    21996
  )
)
```

Q1. Make a plot of `year` vs `cases`

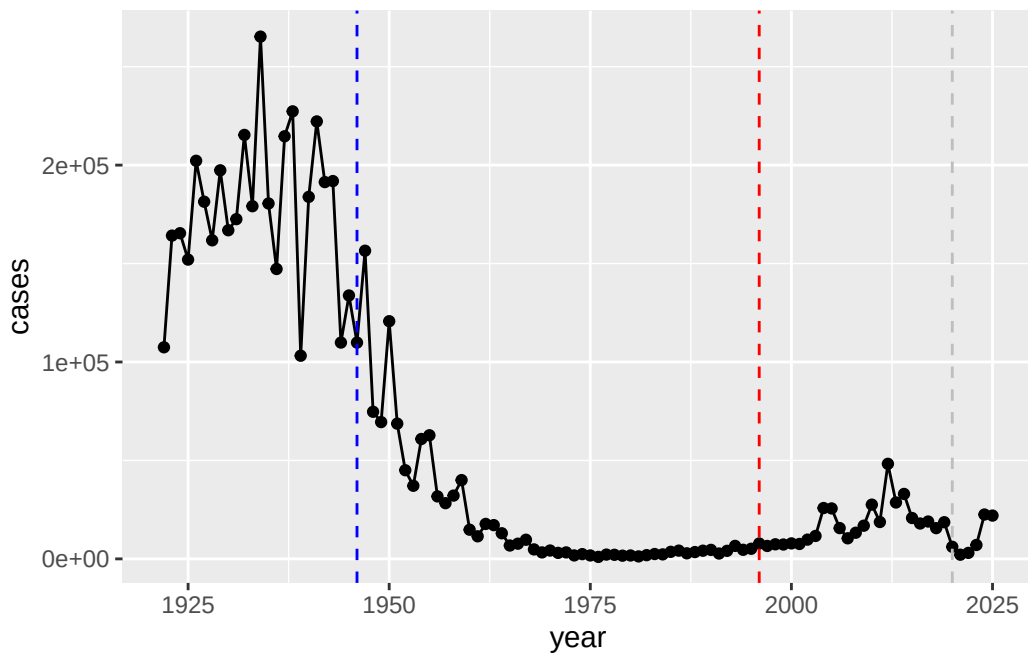
```
library(ggplot2)
ggplot(cdc) +
  aes(x = year, y = cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases by Year (1922-2025)",
```

```
       x = "Year",
       y = "Number of cases")
```



Q2. Add some major milestones including the first wP vaccine rool-out (1946), the switch to the newer aP vaccine (1996).

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty=2) +
  geom_vline(xintercept = 1996, col="red", lty=2) +
  geom_vline(xintercept = 2020, col="gray", lty=2)
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

There were high case numbers in the pre-1940s before the introduction of the wP vaccine, but cases dropped dramatically after the vaccine was introduced. However, after the switch to the aP vaccine, pertussis cases began to increase again. One possible explanation is that the protection from the aP vaccine may wear off faster than the protection from the older wP vaccine.

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB project

Computational Models of Immunity - Pertussis Boost project aims to provide the scientific community with this very information.

They make there data available via JSON format returning API. We can read this in R with the `read_json()` function from the **jsonlite** package:

```r
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```r
head(subject, 3)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)? In terms of race and gender is this dataset representstive of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

```r
library(lubridate)
```

```
Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```r
today()
```

```
[1] "2026-03-07"
```

Q. How many days have passed since new year 2000

```r
today() - ymd("2000-01-01")
```

```
Time difference of 9562 days
```

Q. What is this in years?

```r
time_length( today() - ymd("2000-01-01"),  "years")
```

```
[1] 26.17933
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```r
subject$age <- today() - ymd(subject$year_of_birth)
```

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      27      28      28      29      35 
```

```r
wp <- subject %>% filter(infancy_vac == "wP")
round(summary(time_length(wp$age, "years")))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      33      35      37      40      58 
```

Q8. Determine the age of all individuals at time of boost?

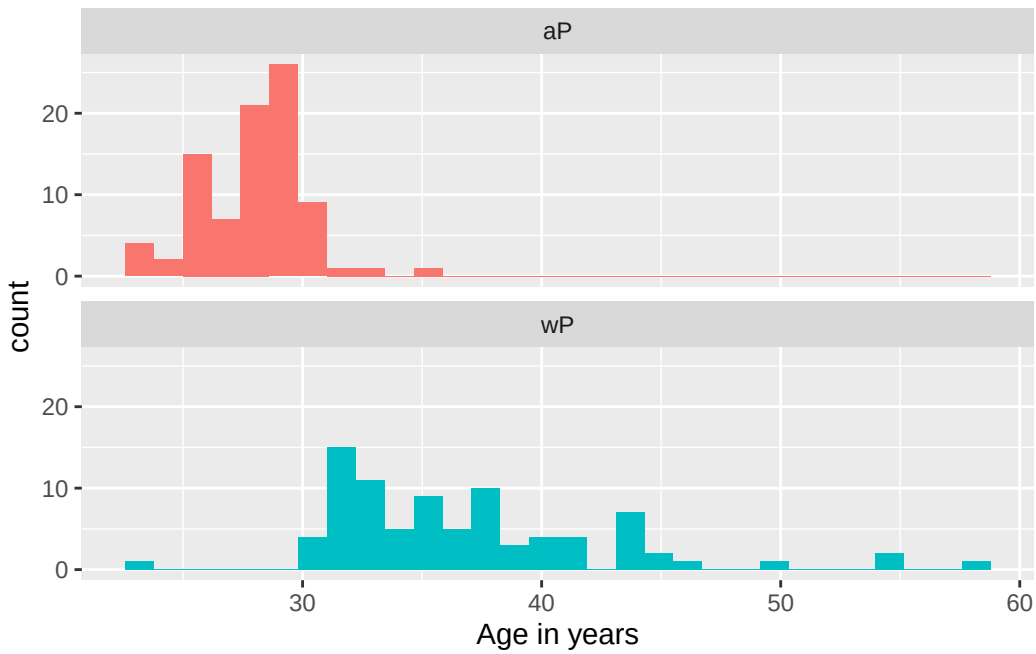
```r
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```r
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Let's read some more database tables:

```r
specimen <- read_json("https://www.cmi-pb.org/api/specimen",
                      simplifyVector = TRUE)

titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer",
                   simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                            -3
```

```
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```
head(titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can the `*_join()` family of functions from **dplyr** to do this

> Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
library(dplyr)
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
dim(meta)
```

```
[1] 1503   14
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset        age specimen_id
1    1986-01-01    2016-09-12 2020_dataset 14675 days           1
2    1986-01-01    2016-09-12 2020_dataset 14675 days           2
3    1986-01-01    2016-09-12 2020_dataset 14675 days           3
4    1986-01-01    2016-09-12 2020_dataset 14675 days           4
5    1986-01-01    2016-09-12 2020_dataset 14675 days           5
6    1986-01-01    2016-09-12 2020_dataset 14675 days           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```r
abdata <- inner_join(titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
dim(abdata)
```

```
[1] 61956    21
```

Q11. What Antibody isotypes are measured for these patients?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q12. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Let's focus on the IgG antigen and make a plot of MFI_normalized for all antigens.

```r
igg <- abdata |>
  filter(isotype=="IgG")

head(igg)
```

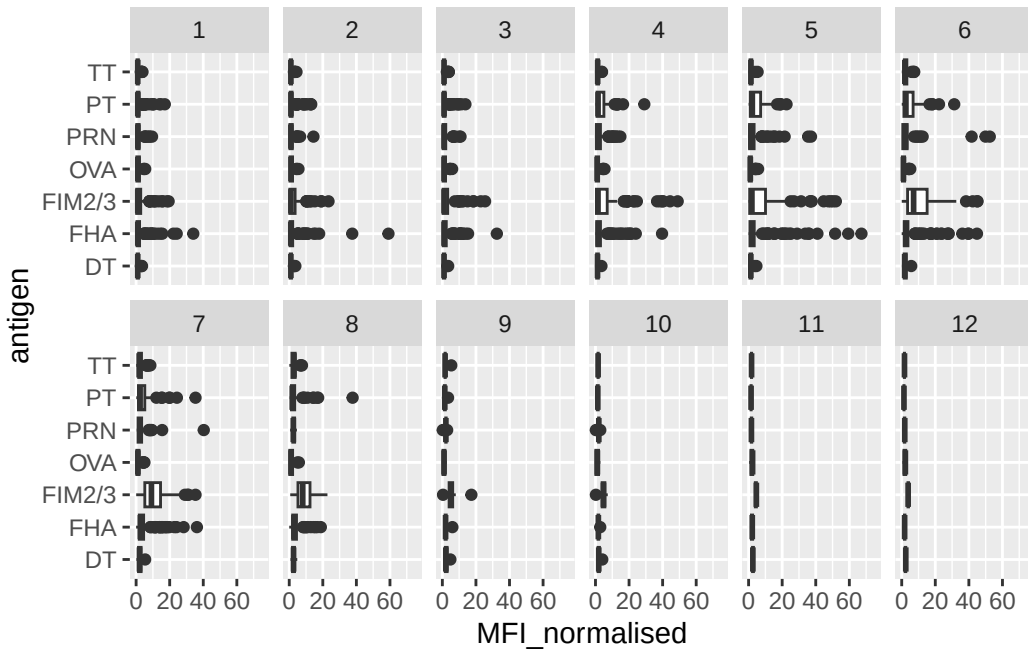
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
```

```
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
         age actual_day_relative_to_boost planned_day_relative_to_boost
1 14675 days                           -3                             0
2 14675 days                           -3                             0
3 14675 days                           -3                             0
4 15771 days                           -3                             0
5 15771 days                           -3                             0
6 15771 days                           -3                             0
  specimen_type visit
1         Blood     1
2         Blood     1
3         Blood     1
4         Blood     1
5         Blood     1
6         Blood     1
```

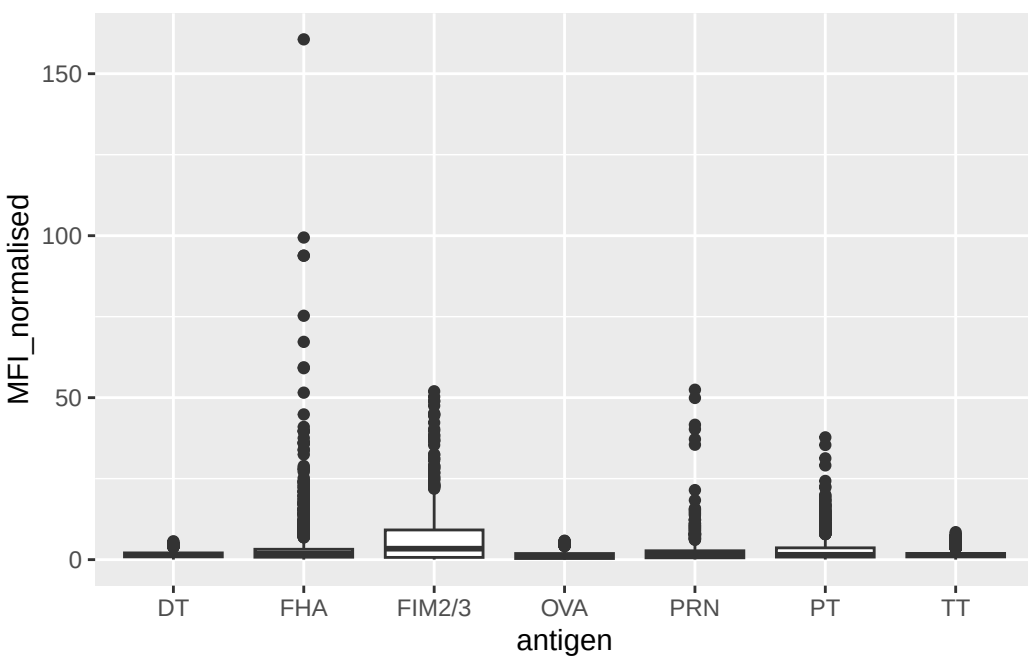
Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
    xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

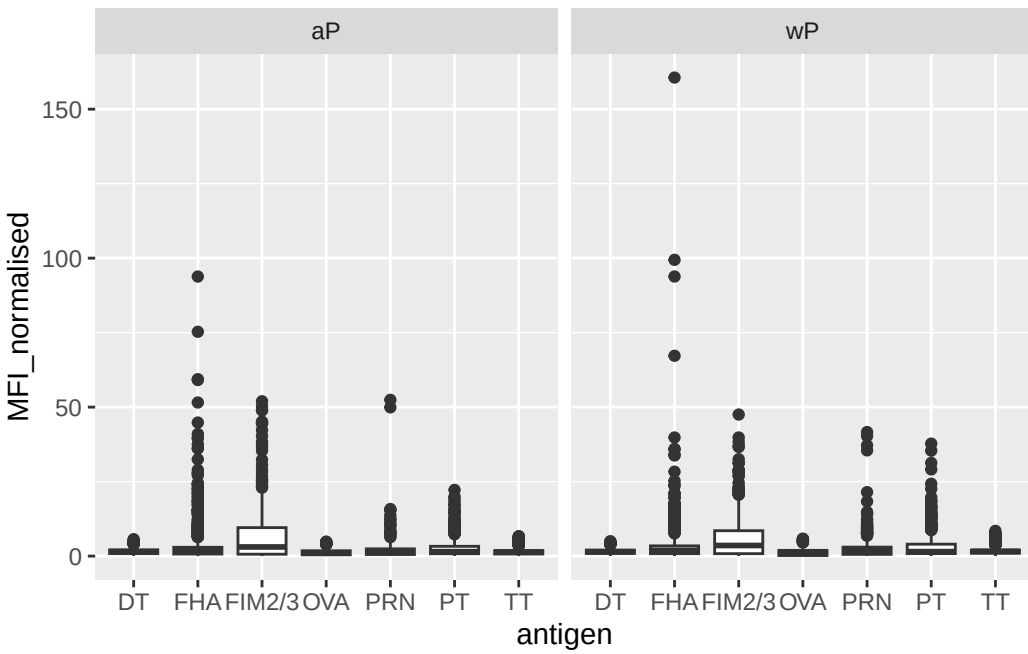



```
ggplot(igg) +
  aes(x = antigen, y = MFI_normalised) +
  geom_boxplot()
```



Q. Is there a difference for aP vs wP individuals with these values?

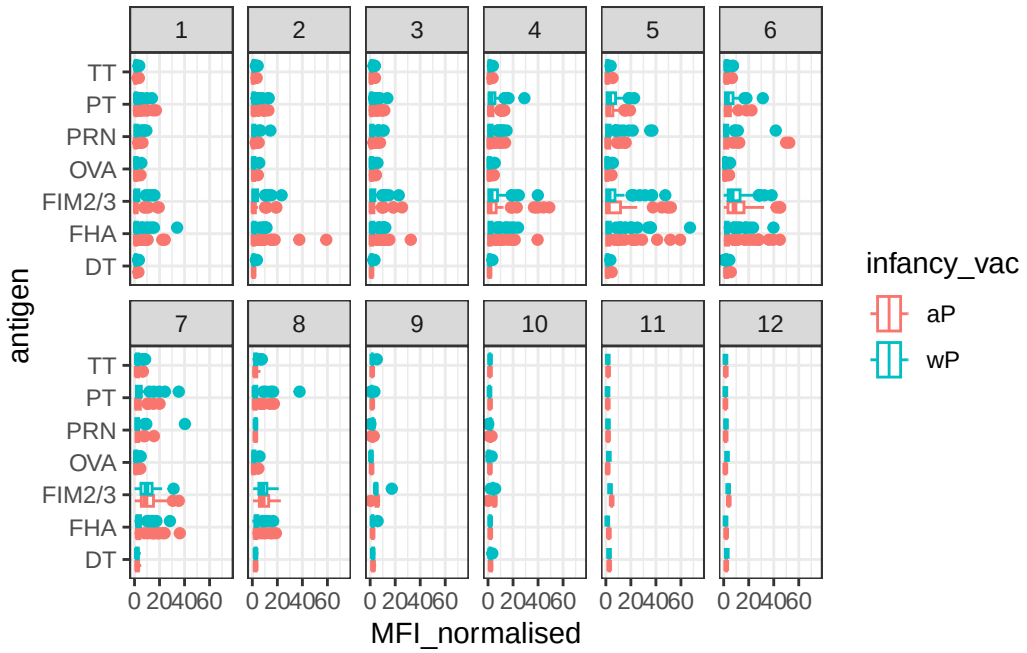
```
ggplot(igg) +
  aes(x = antigen, y = MFI_normalised) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Q14. What antigens show differences in the level of IgG antibody titers recognizing them voer time? Why these and not others?

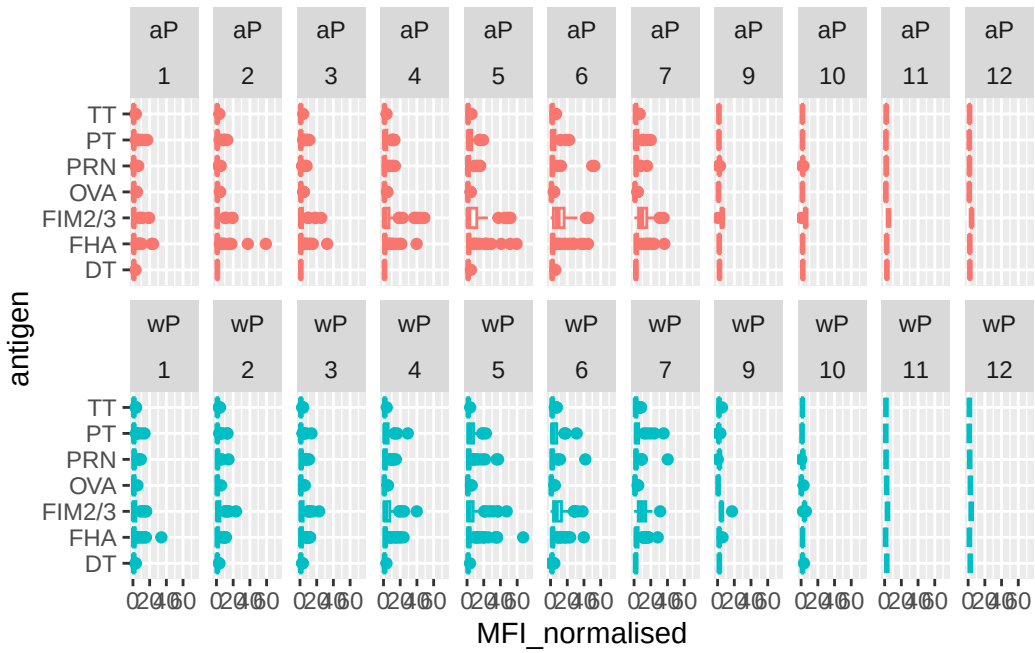
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



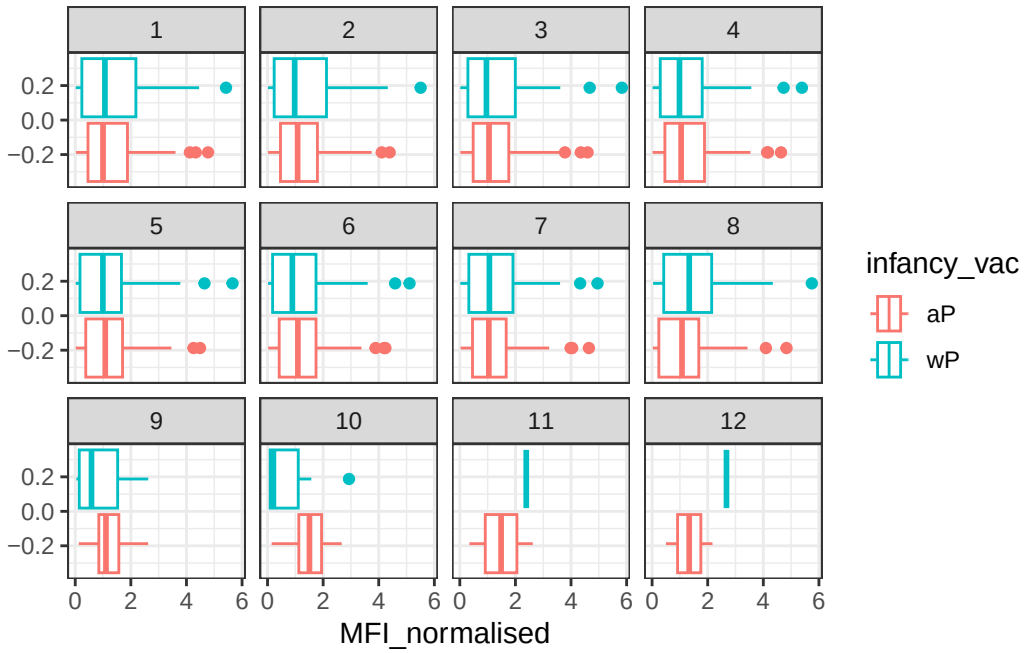

```
igg |> filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

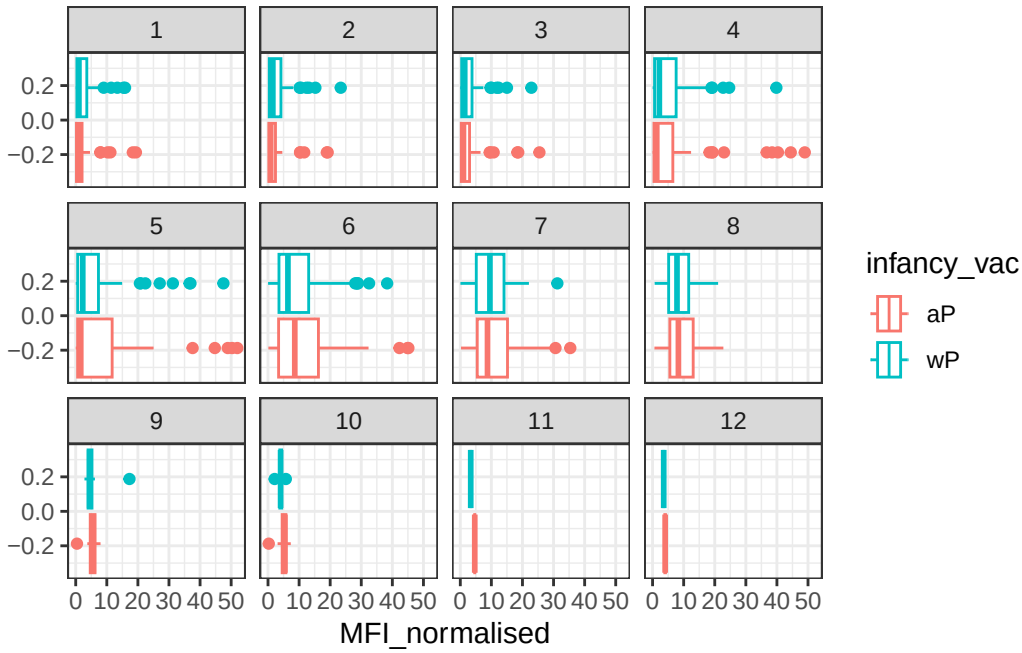


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a "control" antigen ("OVA", that is not in our vaccines) and a clear antigen of interest ("PT", Pertussis Toxin, one of the key virulence factors produced by the bacterium B. pertussis).

```
filter(igg, antigen=="OVA") |>
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```




```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The PT antigen shows a strong increase in IgG levels over time, while the OVA control antigen remains low and relatively constant. PT levels rise after vaccination, peak around visit 5, and then begin to decline. This pattern is similar for both wP and aP vaccinated individuals.

Q17. Do you see any clear difference in aP vs. wP responses?

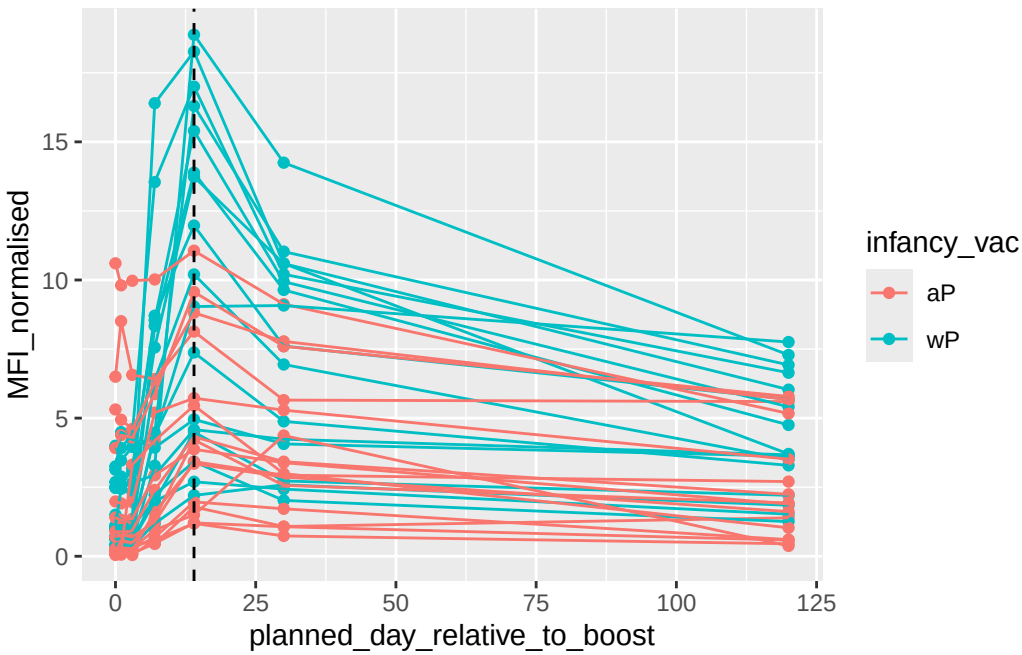
There does not appear to be a clear difference between the aP and wP groups in their responses. Both groups show a similar pattern where PT antibody levels increase over time, peak around visit 5, and then decline.

Focus on "PT" Pertusisis Toxin antigen

```
pt.igg.21 <- igg |> filter(antigen == "PT",
               dataset == "2021_dataset")
```

```
ggplot(pt.igg.21) +
      aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
```

```
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=14, lty=2)
```



Q18. Does this trend look similar for the 2020 dataset?

Yes, the 2020 dataset shows a broadly similar trend. In both years, the response increases sharply from earlier timepoints, reaches a high peak, and then shows a modest decline afterward. The specific numbers differ, but the shape of the pattern is consistent

Obtaining CMI-PB RNASeq data

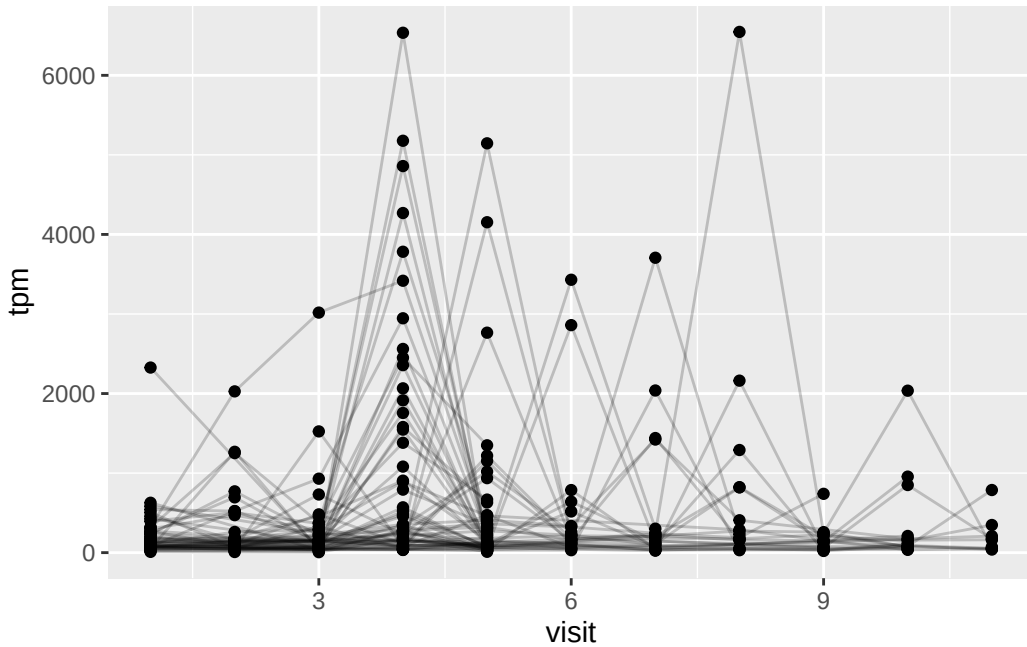
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSG00000211896.7"

rna <- read_json(url, simplifyVector = TRUE)
```

```
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(x=visit, y=tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



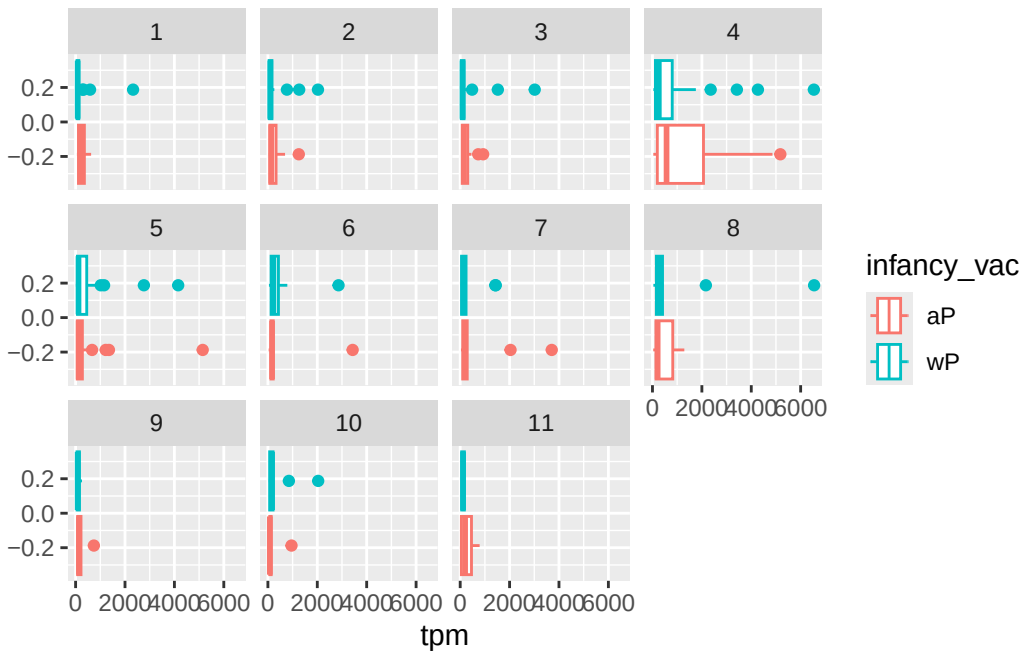
Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

The gene's expression reaches its maximum level around visit 4, and also a similarly high spike at visit 8. However, the single highest TPM value in the entire plot occurs at visit 4.

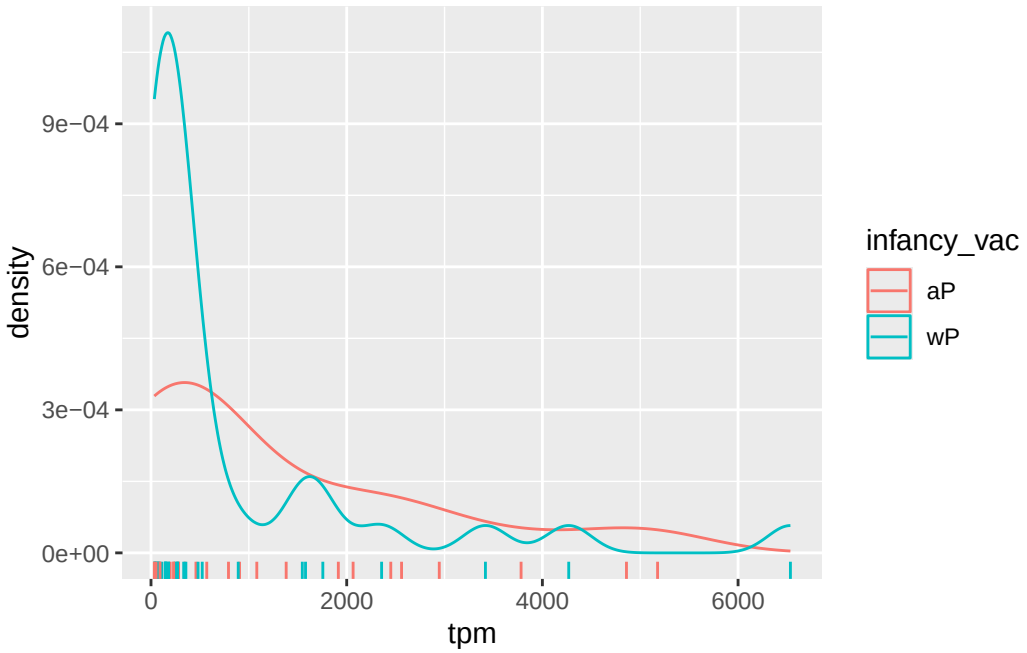
Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Antibody titers show a smooth rise to a peak and then a gradual decline, but the gene expression displays sharp, irregular spikes at specific visits and large variability between individuals.

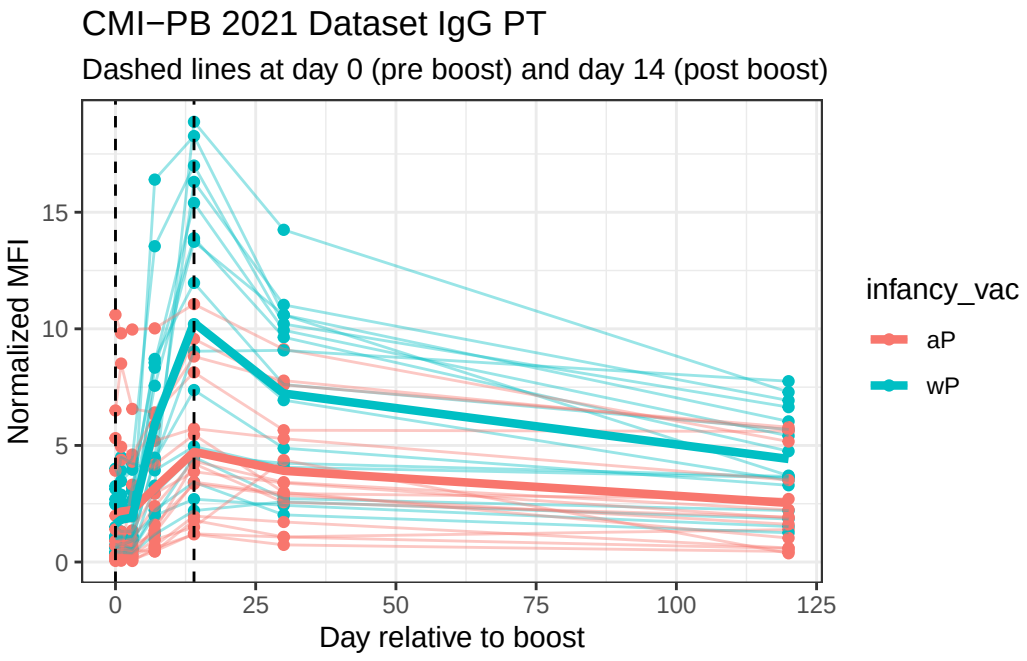
```r
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```




```r
ssrna |>
  filter(visit==4) |>
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```




```
ggplot(pt.igg.21) +
  aes(x = planned_day_relative_to_boost,
      y = MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.4) +
  stat_summary(aes(group = infancy_vac),
               fun = mean,
               geom = "line",
               linewidth = 1.5) +
  geom_vline(xintercept = 0, lty = 2) +
  geom_vline(xintercept = 14, lty = 2) +
  labs(
    title = "CMI-PB 2021 Dataset IgG PT",
    subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
    x = "Day relative to boost",
    y = "Normalized MFI"
  ) +
  theme_bw()
```


CMI−PB 2021 Dataset IgG PT
Dashed lines at day 0 (pre boost) and day 14 (post boost)
Normalized MFI
0
5
10
15
0
25
50
75
100
125
Day relative to boost
infancy_vac
aP
wP